



Invest at the level you're ready for.

A robo advisor that checks what you understand and what losses you can survive before it recommends anything.

Team Lakers · Daniel Gallinas · Martin Christian Herisson · Vasileios Konstantakopoulos · Greta Allison Perbellini

lakersteam.streamlit.app

People do not fail at picking funds. They fail at holding them.

THE BEHAVIOR PROBLEM

Panic selling

When losses pass what an investor can truly tolerate, they sell at the bottom and lock in the damage. The real risk is not volatility. It is the fall that makes you quit.

THE INDUSTRY PROBLEM

Complexity on demand

Modern investing apps hand leveraged products and crypto to anyone who taps a button, whether or not they understand what a 2x fund does on a bad day.

THE RESULT

Portfolios people abandon

Good funds, wrong owner. Sage Invest exists to match the portfolio to the person, with evidence.

Our user, and the one decision the app answers

THE USER

A European retail investor opening their first, or next, investment account.

THE DECISION

"Which portfolio should I buy, given what I actually understand and what losses I can actually sit through?"

Every screen, chart and number in the app serves this single decision.
Nothing decorative survived.

Two scores, kept separate, like real advisors must

APPROPRIATENESS · WHAT YOU MAY BUY

Three knowledge and experience questions gate the product universe. Appetite never unlocks products; understanding does.



Beginner

6 broad index funds



Intermediate

+ Nasdaq 100, property, small caps



Experienced

+ Bitcoin ETC, 2x leveraged fund

SUITABILITY · HOW MUCH RISK FITS YOU

Four questions on horizon, losses, capacity and goal set your loss limit: the worst fall you could hold through without selling.

Cautious –10%

Balanced –15%

Growth –25%

Aggressive –35%

Money needed within 3 years is capped at Cautious, however brave the answers.

This mirrors MiFID II, the EU rule every real broker follows. We made the regulation the product.

Try it while we talk

lakersteam.streamlit.app

Seven questions, two minutes, then your portfolio with the evidence behind it. Live on Streamlit Community Cloud.

If the room wifi dies, we demo from the fallback captures in the repo.



Scan to open the app

The engine: transparent by design

- 1 10,000 candidate allocations.** Random, long only, at most 40% in any single fund, drawn over the user's unlocked universe.
- 2 The sleep test.** Keep only the allocations whose worst fall in 12 years of real history stayed inside the user's loss limit, through COVID and the 2022 bond crash.
- 3 Best survivor wins.** Among portfolios the user could have held, recommend the one that made the most money.
- 4 Always benchmarked.** Splitting money equally is judged by the same test. Sometimes it passes; for many profiles it falls beyond their limit, and the app says so.

Why sampling and not only an optimizer: a drawdown cap is not a convex constraint, and sampling shows the whole feasible landscape. A convex optimizer (skfolio) runs as an independent check.

Evidence, not vibes

RECOMMENDED (BEGINNER, -15%
LIMIT)

8.4% per year

Worst episode **-14.6%**

within the limit

SPLITTING EQUALLY (1/N)

7.3% per year

Worst episode **-20.6%**

beyond the limit

€50,000 BECAME

€188,759

2014 to 2026, twelve years of daily
data

WE TESTED OUR OWN HONESTY

Choosing the portfolio on the first six years only, the pick delivered **8.7% per year** in sample but **6.5% per year** on the unseen half, with a worst fall of **-17.4%**, past the limit. That gap is what picking winners on history costs. The app shows it to every user instead of hiding it.

Trust, but verify. Then verify again.

INDEPENDENT RECOMPUTATION

verify.py

Recomputes return, volatility and worst fall from raw prices with code that imports nothing from the app. Agreement to 4 decimals required. During development it caught a real windowing bug. That is why it exists.

AUTOMATED END TO END TESTS

test_wizard.py

Drives the full app headlessly for three personas, including the edge case where no portfolio can meet the limit and the app must explain instead of crash.

SECOND OPINION

skfolio cross check

A convex optimizer solves a relaxation of the same problem. At low risk it lands where we land, validating the sampling. At high risk it piles funds at their caps, exactly the corner solutions we chose not to present as advice.

Data you can audit

PROVENANCE

Yahoo Finance daily adjusted closes for 11 EUR listed UCITS funds on Euronext Amsterdam and Xetra. Retrieved live each session; a frozen snapshot ships with the repo as fallback and as the deterministic input for verification.

CLEANING, EVIDENCED

Common window per universe. Exchange holiday gaps carried forward up to 3 days; longer gaps dropped, never invented. The app displays the exact counts: cells filled and dates dropped for your universe.

COVERAGE, DISCLOSED

The riskiest tier only has data since June 2020, a window that never saw the COVID crash. The app warns that user in plain words rather than letting a short window flatter risky assets.

What we would not let you miss

HONEST LIMITATIONS

Past drawdowns understate the future: this window holds no 2008. Costs and taxes are excluded. Picking the best of 10,000 is selection on one history, and we measure that cost out of sample. The projection reshuffles the past; it cannot imagine a new kind of crisis.

AI USE, DISCLOSED

Built with Claude Code as pair programmer under the team's product direction. Every headline number is recomputed by independent, non AI code before we trust it. The in app chat runs on OpenAI, grounded on the user's own numbers, educational only. Full disclosure in the repo.

Ask Sage: the advisor explains itself

"Is -22.9% a lot? What does it mean for me?"

"A drawdown of -22.9% means that, at its worst point in the past, this portfolio dropped by that much from its peak. With your -25% limit it is significant but within what you said you could hold..."

Grounded. The chat knows this user's tier, loss limit, weights and results, so answers are about their portfolio, not generic finance.

Bounded. Educational answers only, never advice, honest about the method's limits.

Optional. Without an API key the chat simply does not exist, and nothing else depends on it.

Where this goes next

NEXT

A cash tier

Short horizons deserve money market funds, not a warning.

NEXT

Rebalancing advisor

Our backtest says yearly is enough; see the appendix.

NEXT

Suitability reports

Exportable MiFID style documentation of every recommendation.

WHO OWNS WHAT

Daniel · product and demo

Martin · data pipeline

Vasileios · method and metrics

Greta · verification and limitations

Thank you. Questions welcome.

lakersteam.streamlit.app

Appendix: how often should you rebalance?

Same recommended portfolio, five maintenance policies, twelve years of history.

POLICY	RETURN PER YEAR	WORST FALL	VERDICT
Never rebalance	9.4%	-15.5%	drifts past the -15% limit
Monthly	8.3%	-14.5%	safe, extra effort
Quarterly	8.4%	-14.1%	safe
Yearly	8.3%	-14.1%	safe, our advice
On 5 point drift	8.5%	-14.4%	safe, our advice

Rebalancing is risk control, not extra return. Skipping it looks better on paper only because the mix quietly drifts riskier than the user agreed to. Yearly, or when any fund drifts five points from target, captures the benefit.